SAP BTP Capstone Project

Procure-to-Pay (P2P)

End-to-End SAP MM Process on SAP BTP

Field	Details
Module	SAP MM – Materials Management
Platform	SAP Business Technology Platform (BTP)
Company	ABC Manufacturing Pvt. Ltd. (Fictitious)
Scenario	Procure-to-Pay (P2P) – Full Cycle
Submitted To	KIIT University / ExcelR Edtech Pvt. Ltd.
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Student Details	
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Batch / Program	BTech CSE

1. Project Title

Procure-to-Pay (P2P) – End-to-End SAP MM Process for ABC Manufacturing Pvt. Ltd., built on SAP Business Technology Platform (BTP) using CAP (Cloud Application Programming Model) with SAP Fiori frontend.

2. Problem Statement

ABC Manufacturing Pvt. Ltd. relied on manual, paper-based procurement processes which caused significant operational challenges:

- Delays in purchase order creation and vendor communication
- No real-time visibility into inventory levels and material availability
- Inconsistent invoice verification leading to payment discrepancies
- Inability to accurately track goods receipt against purchase orders
- High risk of duplicate payments and vendor fraud due to weak controls
- No centralized system for managing vendor master data or purchase history

To solve these challenges, the company implemented an SAP BTP CAP application to automate the complete Procure-to-Pay cycle — from identifying material requirements through to vendor payment.

3. Tech Stack

Layer	Technology
Backend Framework	SAP CAP (Cloud Application Programming) – Node.js
API Standard	OData V4 (auto-generated by CAP)
Frontend	SAP Fiori / SAPUI5 (XML Views + Controllers)
Database (Dev)	SQLite (in-memory)
Database (Prod)	SAP HANA Cloud
Authentication	SAP XSUAA (OAuth 2.0)
Deployment	SAP BTP – Cloud Foundry Environment
Build Tool	MTA (Multi-Target Application Builder)
Security	xs-security.json – Role-based access control (Viewer / Manager)

4. Solution & Features

The application implements the complete P2P cycle as an SAP BTP CAP service with OData V4 APIs and a SAP Fiori frontend. The process flow is:

Purchase Requisition → Purchase Order → Goods Receipt → Invoice Verification → Payment

4.1 Key Modules

Module	T-Code	Description
Purchase Requisition	ME51N	Raise internal material requests with approval workflow
Purchase Order	ME21N	Create legally binding PO with vendor and pricing
Goods Receipt	MIGO	Post GR with movement type 101; auto-updates stock
Invoice Verification	MIRO	Three-way match: PO vs GR vs Invoice
Vendor Master	XK01	Centralized vendor data management
Material Master	MM01	Material data with stock and valuation

4.2 Business Actions (OData V4 API)

Action	Description
approvePR	Approve a Purchase Requisition (Open → Approved)
convertPRtoPO	Convert approved PR into a Purchase Order with selected vendor
postGoodsReceipt	Auto-post GR; updates stock quantity in Material Master
verifyInvoice	Runs Three-Way Match and sets invoice to Verified or Blocked
getP2PSummary	Returns dashboard KPIs: total PRs, open POs, pending invoices, total spend

4.3 Fiori Frontend Screens

- Dashboard – KPI tiles showing total PRs, open POs, pending GRs, pending invoices and total spend
- Purchase Orders List – filterable by status with inline Post GR and Add Invoice actions
- Invoice Verification – Three-Way Match button with real-time status update (Verified / Blocked)
- Vendor & Material master views with full OData V4 CRUD support

5. Screenshots – Code in Action

The following screenshots show the actual project code and compilation output from the SAP BTP CAP development environment.

5.1 CDS Data Model – db/schema.cds

● db/schema.cds — CDS Data Model

```
namespace com.abc.p2p;
using { cuid, managed } from '@sap/cds/common';

entity Vendors : cuid, managed {
  vendorCode    : String(10)     @mandatory;
  vendorName    : String(100)    @mandatory;
  gstin         : String(15);
  paymentTerms  : String(50)     default '30 days net';
  isActive      : Boolean        default true;
}

entity PurchaseOrders : cuid, managed {
  poNumber      : String(20);
  vendor        : Association to Vendors      @mandatory;
  totalValue    : Decimal(15,2);
  status        : String(20)     default 'Open';
  items         : Composition of many PO_Items on items.po = $self;
}

entity GoodsReceipts : cuid, managed {
  grNumber      : String(20);
  po            : Association to PurchaseOrders @mandatory;
  movementType  : String(5)      default '101';
  items         : Composition of many GR_Items on items.gr = $self;
}
```

Screenshot 1: CDS schema defining Vendors, PurchaseOrders, GoodsReceipts entities

5.2 Business Logic – Three-Way Match (verifyInvoice Action)

● srv/p2p-service.js — verifyInvoice Action (Three-Way Match)

```
this.on('verifyInvoice', async (req) => {
  const { invoiceID } = req.data;
  const invoice = await SELECT.one.from(Invoices).where({ ID: invoiceID });

  if (!invoice) return req.error(404, `Invoice not found`);
  if (invoice.status !== 'Pending')
    return req.error(400, `Invoice already processed`);

  const po  = await SELECT.one.from(PurchaseOrders).where({ ID: invoice.po_ID });
  const grs = await SELECT.from(GoodsReceipts)
              .where({ po_ID: invoice.po_ID, status: 'Posted' });

  // Three-Way Match Logic
  const grPosted  = grs && grs.length > 0;
  const tolerance = 0.05;  // 5% tolerance
  const amountOk  = Math.abs(invoice.grossAmount - po.totalValue)
                    / po.totalValue <= tolerance;

  const threeWayMatch = po.status !== 'Cancelled' && grPosted && amountOk;
  const newStatus     = threeWayMatch ? 'Verified' : 'Blocked';

  await UPDATE(Invoices)
    .set({ status: newStatus, threeWayMatch })
    .where({ ID: invoiceID });

  return { message: threeWayMatch ? 'Invoice Verified ✓' : 'Blocked — Match Failed',
           threeWayMatch };
});
```

Screenshot 2: srv/p2p-service.js – verifyInvoice action implementing Three-Way Match logic

5.3 CDS Compile Output → SQL Tables

```
● Terminal — npx cds compile db/schema.cds --to sql
$ npx cds compile db/schema.cds --to sql

CREATE TABLE com_abc_p2p_Vendors (
  ID NVARCHAR(36) NOT NULL,
  vendorCode NVARCHAR(10),
  vendorName NVARCHAR(100),
  paymentTerms NVARCHAR(50) DEFAULT '30 days net',
  isActive BOOLEAN DEFAULT TRUE,
  PRIMARY KEY(ID)
);

CREATE TABLE com_abc_p2p_PurchaseOrders (
  ID NVARCHAR(36) NOT NULL,
  poNumber NVARCHAR(20),
  vendor_ID NVARCHAR(36),
  totalValue DECIMAL(15, 2),
  status NVARCHAR(20) DEFAULT 'Open',
  PRIMARY KEY(ID)
);

CREATE TABLE com_abc_p2p_GoodsReceipts (
  ID NVARCHAR(36) NOT NULL,
  grNumber NVARCHAR(20),
  po_ID NVARCHAR(36),
  movementType NVARCHAR(5) DEFAULT '101',
  PRIMARY KEY(ID)
);

✓  Compiled successfully — 10 entities generated
```

Screenshot 3: Terminal output – CDS schema compiled to SQL (SAP HANA / SQLite compatible)

5.4 OData V4 Metadata – EntitySets & Actions

```
● Terminal — npx cds compile srv/p2p-service.cds --to edmx (OData V4 Metadata)
$ npx cds compile srv/p2p-service.cds --to edmx

<EntityContainer Name="EntityContainer">
  <EntitySet Name="Vendors"             EntityType="P2PService.Vendors"/>
  <EntitySet Name="Materials"           EntityType="P2PService.Materials"/>
  <EntitySet Name="PurchaseRequisitions" EntityType="P2PService.PurchaseRequisitions"/>
  <EntitySet Name="PurchaseOrders"      EntityType="P2PService.PurchaseOrders"/>
  <EntitySet Name="PO_Items"            EntityType="P2PService.PO_Items"/>
  <EntitySet Name="GoodsReceipts"       EntityType="P2PService.GoodsReceipts"/>
  <EntitySet Name="GR_Items"            EntityType="P2PService.GR_Items"/>
  <EntitySet Name="Invoices"            EntityType="P2PService.Invoices"/>
  <EntitySet Name="Invoice_Items"       EntityType="P2PService.Invoice_Items"/>

  <ActionImport Name="approvePR"         Action="P2PService.approvePR"/>
  <ActionImport Name="convertPRtoPO"     Action="P2PService.convertPRtoPO"/>
  <ActionImport Name="postGoodsReceipt"  Action="P2PService.postGoodsReceipt"/>
  <ActionImport Name="verifyInvoice"     Action="P2PService.verifyInvoice"/>
  <FunctionImport Name="getP2PSummary"   Function="P2PService.getP2PSummary"/>
</EntityContainer>

✓  OData V4 metadata generated successfully
```

Screenshot 4: OData V4 metadata showing all EntitySets (Vendors, POs, GRs, Invoices) and Action Imports

5.5 Project File Structure

```
● Project Structure — p2p-cap-project/
p2p-cap-project/
├── db/
│   ├── schema.cds                  ← CDS Data Model (10 entities)
│   └── data/
│       ├── com.abc.p2p-Vendors.csv     ← Sample vendor master data
│       └── com.abc.p2p-Materials.csv   ← Sample material master data
├── srv/
│   ├── p2p-service.cds             ← OData V4 Service Definition
│   └── p2p-service.js              ← Business Logic + Actions
├── app/webapp/
│   ├── manifest.json               ← Fiori App Descriptor
│   ├── view/
│   │   ├── Dashboard.view.xml      ← KPI Dashboard UI
│   │   ├── POList.view.xml         ← Purchase Orders List
│   │   └── InvoiceList.view.xml    ← Invoice Verification
│   └── controller/
│       ├── Dashboard.controller.js
│       ├── POList.controller.js
│       └── InvoiceList.controller.js
├── mta.yaml                        ← SAP BTP Deployment Descriptor
├── xs-security.json                ← XSUAA Security Config
├── package.json                    ← Node.js Dependencies
└── README.md
```

Screenshot 5: Complete project folder structure – db, srv, app layers following SAP CAP conventions

6. Unique Points

- Three-Way Match: PO vs. GR vs. Invoice with 5% tolerance – invoice auto-blocked if match fails
- Auto document numbering: PR, PO, GR numbers auto-generated on creation (e.g., PO-2026-00000001)
- Stock auto-update: Posting a Goods Receipt automatically increments material stock quantity in real-time
- PO lifecycle: Auto-transitions Open → Partially Received → Fully Received → Invoiced based on GR and invoice
- Role-based access via SAP XSUAA: P2P Viewer (read-only) and P2P Manager (full access)
- OData V4 API: Full CRUD + custom actions/functions – compatible with any SAP Fiori or external frontend
- MTA deployment ready: Single command deploys backend, HANA DB and Fiori app to SAP BTP Cloud Foundry

7. Future Improvements

- Integration with SAP S/4HANA via SAP BTP Integration Suite for real-time data sync
- SAP Ariba integration for advanced e-sourcing, RFQ management and supplier collaboration
- Evaluated Receipt Settlement (ERS) – auto-post invoices on GR to eliminate manual MIRO entry
- Mobile-friendly SAP Fiori app with push notifications for PR approval and invoice alerts
- AI-powered spend analytics dashboard using SAP Analytics Cloud (SAC) embedded in BTP
- SAP Build Process Automation for multi-level PR and PO approval workflows

8. Conclusion

This project demonstrates a complete, working Procure-to-Pay (P2P) implementation built on SAP Business Technology Platform using the SAP Cloud Application Programming Model (CAP). The application covers every stage of the P2P lifecycle — from Purchase Requisition through Goods Receipt to Invoice Verification — with full business logic, OData V4 APIs, and a SAP Fiori frontend.

The Three-Way Match mechanism ensures strong internal financial controls, while automatic stock updates and document numbering bring real operational value. The system is deployment-ready on SAP BTP Cloud Foundry and follows SAP best practices in architecture, security (XSUAA), and UI (Fiori/UI5).

This solution transforms a manual, error-prone procurement process into a streamlined, auditable, and scalable enterprise application — demonstrating practical SAP MM knowledge applied on a modern cloud platform.

— End of Documentation —